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A market resilient data-driven approach to option pricing

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In this paper, we present a data-driven ensemble approach for option price prediction whose derivation is based on the no-arbitrage theory of option pricing. Using the theoretical treatment, we derive a common representation space for achieving domain adaptation. Through a specific scaling, suitable for financial time series data, we obtain a feature representation that is indistinguishable for samples coming from different domains. This provides an advantage over conventional models when predicting atypical out-of-sample test data. The success of an implementation of this idea is shown using some real market data. The root mean squared error in prediction turns out to be less than one-third of that for the benchmark model. We further report several experimental results for critically examining the predictive performance of the derived pricing models.

Keywords: Option pricing; Computational finance; Non-parametric approach; Machine learning; Domain adaptation

1. Introduction

Some products or services are often bought to protect the buyer from uncertainties. Warranty is one of them. Figuring out a fair price for a warranty is not an easy task. Since the financial markets consist of risky assets, traders purchase some warranty-type contracts, called options. Several different types of options are traded by numerous traders daily in every modern stock exchange. Determination of an option's fair price is one of the central questions in Mathematical Finance. Following the seminal work (Black and Scholes 1973) by Black and Scholes the mathematical treatment for addressing the option pricing problem quickly flourished. In the last fifty years, thousands of peer-reviewed articles came into existence to address various mathematical, computational, or statistical aspects of fair pricing of a large spectrum of options under diverse scenarios. Consideration of increasingly realistic stochastic models of asset returns and then showing that the model could preclude arbitrage, before deriving a stochastic or PDE representation of an option's price is the state of the art for option pricing research. These works have tremendously enriched our understanding of fair

pricing of options, beyond any doubt. Besides, option pricing research also played a vital role in inspiring deeper and more diverse investigation of stochastic calculus, differential equations, numerical simulations etc. Given this, it is natural to have a judgment that 'pricing of option is not just about data science'. Without disagreeing with that judgment, we propose a novel direction of research that translates some existing data science ideas into the language of stochastic modeling to derive some theoretical algorithms of data-driven option pricing that leverage from both worlds.

A solution to the option pricing problem is called data-driven if the approach assumes no theoretical model of asset dynamics and relies only on the observed data. However, it is possible to utilize the general theory of fair pricing of options on an abstract underlying asset while deriving a data-driven option pricing model. In other words, despite the pricing solution being data-driven the derivation of the solution need not always be data-driven. For example, in a recent work (Cao *et al.* 2021), some theoretical constraints of fair price of options are utilized to come up with a data-driven model. The investigations on the prospect of various types of data-driven option pricing are not new in the literature. Nevertheless, the volume of work in this direction is much less than its counterpart in the theoretical option pricing. A comprehensive literature survey is beyond the scope of this paper. In the initial works of Malliaris and Salchenberger (1993), Hutchinson

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et al. (1994) and Qi and Maddala (1996) the promise of data-driven option pricing is reported using the data of option contracts on the S&P 500. Further investigation on feature selection appeared in Boek *et al.* (1995) and Malliaris and Salchenberger (1996). Modularity is used in Gradojevic *et al.* (2009) and Yao *et al.* (2000) for improving the pricing quality. In the context of option pricing, modularity entails dividing the data set into disjoint modules, according to the moneyness and time-to-maturity parameters of the contracts. Some more recent works (Das and Padhy 2017, Goswami *et al.* 2021, Shubham *et al.* 2023, Sharma *et al.* 2025) build upon the above findings and a theoretical property of options contracts, called homogeneity hint (Garcia and Gençay 2000). While homogeneity hint implies that the ratio of option and spot prices is a suitable scale-free target, the suitability of implied volatility (IV) as another scale-free target is also undeniable (Cao *et al.* 2020). Often the scarcity of data limits the application of data-driven solution to the option pricing problem. Multi-source training, which is not uncommon in finance applications (Hoseinzade and Haratizadeh 2019), provides a good remedy. An implementation of multi-source training for option pricing appears in Goswami *et al.* (2021). Although this allows a certain degree of generalization, but lacks reasonable domain adaptation. To the best of our knowledge, a thorough investigation of the domain adaptability of data-driven solutions to the option pricing problem is missing from the literature. In this paper, we take a step toward addressing this gap by deriving a common representation space that enables effective domain adaptation in data-driven option pricing. We cite Sharma and Kumar Verma (2025) and the references therein for a more comprehensive literature survey on the data-driven solution to the option pricing problem. It is also worth noting that since we do not calibrate a theoretical model for the option price prediction, the discussion of no-arbitrage calibration like Fengler (2009) is irrelevant here. We also do not address the question of hedging options. A comprehensive literature survey on the statistical hedging problem can be found in Ruf and Wang (2020).

In this paper, we first revisit the homogeneity hint property by producing theoretical proof in a wide general setting. Then we explain the shortcomings of the option price prediction models based only on the homogeneity hint. Such models can learn from one asset class data and be applied to another, provided their log returns have similar distributions under their risk-neutral measures. However, requiring this similarity in two different assets is quite restrictive. For this reason, we formulate, in a parametric setting, a common representation space that bridges two different risk-neutral return distributions. The common representation space is used for connecting option prices on assets having no similarities. This is a fresh idea for the machine learning application of option pricing, where domain adaptation is implemented. We show theoretically, as well as empirically that a data-driven approach based on a common representation space can gain accuracy on test data having significant domain shifts. For the empirical study, we consider the option price data of two indices in the National Stock Exchange (NSE) of India mainly because the options on these indices (NIFTY 50, and NIFTY Bank) have very high trading volume in respect of the global market.

We compare the performance of both approaches, the one that is based on homogeneity hint (HH) and the one that accounts for domain shift (DS) using the price data of options on NIFTY 50, and NIFTY Bank indices. We identify that option price data during the COVID lockdown period in India has a significant domain shift and is hence eligible to assess the performance of models for domain shift. The assessment shows a significant gain in accuracy for models that utilize a common representation space. We also propose an ensemble model using the models based on HH and the one that is to address DS. We put forward a notion of domain shift quotient, that measures the degree of domain shift and is utilized to set weights on constituent models in the ensemble model. We have performed extensive experiments to assess the performance of the ensemble model using empirical as well as synthetic data. In this paper, we also assess the performance of each of these three approaches for multi-source training models, where the model is trained on data coming from symbols NIFTY 50 and NIFTY Bank. We report empirical evidence of the reliability of the multi-source ensemble model. Option price prediction model with multi-source training is capable of giving reliable answers even for an underlying asset, which has no or little historical data. This overcomes the common hurdle faced by any data-driven approach.

It is worth mentioning that we include no macroeconomic or fundamental financial variables in the feature sets, to retain the interpretability of the model. Thus the proposed approaches do not lack the interpretability of the model which is a primary criticism that a data science model commonly receives. We also clarify that the goal of this paper is not merely to showcase the usefulness of an existing or newly developed machine-learning method. This rather lays the foundation of the theoretical solution to domain adaptation, a data science idea, in the context of option pricing. The full potential of a non-asset-specific model with the above-mentioned domain adaptation may further be explored by extensive experimentation. In this connection, we recall that Sirignano and Cont (2019) reports a similar extensive experiment to study some other universal non-asset-specific relations captured by a deep learning model. However, keeping in mind that this is the first research on domain adaptation in the context of option pricing, we focus more on theory building than on large-scale empirical verification across all markets.

In this paper, we introduce a variable called *volatility scalar* that plays a key role in the formulation of common representation space. Theoretically, it represents the root mean square of the volatilities of the underlying during the remaining life of the option, provided the underlying follows diffusion dynamics. This variable is used for a specific scaling at the stage of defining features in the common representation space. We derive its expressions under the constant volatility as well as stochastic volatility scenarios. These expressions allow one to estimate the value of the volatility scalar. However, we retain simplicity in the implementation by assuming constant volatility while estimating the variable, and observe that the notion of domain adaptation works even with such a crude assumption.

The rest of the paper is organized as follows. In Section 2, we develop the theoretical framework of approaches based on

HH and one for addressing DS. For achieving the domain adaptation, a common representation space is formulated using an approximate identity in Section 3. In this section, we develop an ensemble approach too. The empirical data is described in Section 4, using exploratory data analysis. Subsequently, we describe all proposed supervised learning approaches in Section 5. In Section 6, we report the performance of every model based on proposed approaches. Some concluding remarks are added to Section 7.

2. Theoretical framework

2.1. Non-parametric settings

Let $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_u\}_{u \geq 0})$ be a filtered probability space. In this section we assume that $S^{(t,s)} := \{S^{(t,s)}(u), u \geq t\}$ is an adapted right continuous with left limit (rcll) process that models the price of a risky asset for all future time $u \geq t$ and at present time $t(\geq 0)$ it is equal to a positive value s . Evidently, $S = \{S^{(t,s)} \mid t \geq 0, s > 0\}$ forms a family of adapted processes. With a minor but usual abuse of terminology, we often call S itself as a process. Let $B := \{B(u)\}_{u \geq 0}$ denote the price process of a risk-free asset. We further assume the following.

Model Assumptions:-

- (M1) The market model, consisting of risky and risk-free assets with price processes $S^{(0,s)}$ and B respectively is free of arbitrage under admissible strategies for any $s > 0$.
- (M2) The family of processes S satisfies

$$S^{(t,s)}(u) = sS^{(t,1)}(u), \quad (1)$$

for all $u \geq t \geq 0$ and $s > 0$ almost surely.

- (M3) Either $S^{(0,s)}$ is Markov or there is an additional auxiliary process $V := \{V(u)\}_{u \geq 0}$ such that $\{(S^{(0,s)}(u), V(u))\}_{u \geq 0}$ is Markov w.r.t. the filtration $\{\mathcal{F}_u\}_{u \geq 0}$ for all $s > 0$.

It is important to note that M1 is a standard technical assumption. On the other hand, M2 asserts affine structure, i.e. S is expressed as exponential. This is valid, for example, with Geometric Brownian motion (GBM), regime-switching GBM, Merton's Jump Diffusion model, etc. Additional V in M3 is not needed for GBM, but that is essential for other multi-factor models. For example, V is the Markov regime in a regime-switching model, instantaneous volatility in the Heston model, etc.

DEFINITION 2.1 (Moneyness) Let K be the strike price of a call option. If the present asset price is s , $p = K/s$ is called the present moneyness of the said option. Note that $p = 1$ represents at-the-money, $p > 1$ represents out-of-the-money and $p < 1$ represents in-the-money.

THEOREM 2.2 Assume that the market contains two risky assets other than a risk-free asset, whose price processes are modeled by S_1 and S_2 , satisfying (M1)–(M3) with a common auxiliary process V . Let φ_i denote the European call option price function on the i th risky asset. That is, $\varphi_i(t, s_i, v; K_i, T)$ gives the present (time t) price of the call option on S_i having

exercise price and maturity K_i and T respectively with $V(t) = v$ and $S_i(t) = s_i$. Also assume that the conditional laws of $S_1^{(t,1)}(T)$ and $S_2^{(t,1)}(T)$ given $V(t) = v$ are identical under their corresponding risk-neutral measures $\mathbb{P}_1$ and $\mathbb{P}_2$, respectively. If the contracts are chosen with an identical time to maturity and equal moneyness at present time t i.e. $\frac{K_1}{s_1} = \frac{K_2}{s_2} = p$ (say), we have

$$s_1^{-1} \varphi_1(t, s_1, v; p s_1, T) = s_2^{-1} \varphi_2(t, s_2, v; p s_2, T). \quad (2)$$

The above theorem is applicable for a wide range of models. For example, when two geometric Brownian motions (GBM) with different drifts but the same diffusion coefficients are taken, their dynamics under respective risk neutral measures are identical. The requirement of common V is also not a real restriction. For example, consider two regime switching GBMs S_1 and S_2 , with Markov regime processes X_1 and X_2 respectively. In other words, for each i , X_i is a finite state continuous-time Markov chain that influences otherwise constant drift and volatility coefficients of a GBM S_i . Then the process $V := (X_1, X_2)$, is such that both (S_1, V) and (S_2, V) are Markov. It is also worth noticing that due to the normalization condition in the equality of conditional law, the theorem remains applicable even if one asset's price is more than several orders of magnitude of another. A proof of this theorem is given in the appendix.

REMARK 2.3 (Homogeneity Hint Approach or $\mathcal{A}_{HH}$) The above theorem is alluded to as *homogeneity hint* (see Garcia and Gençay 2000 and references therein) as the scale-free terms on both sides of (2) are equal. This also reflects the homogeneity of order one of the option prices in stock and strike prices. For predicting option price, the scale-free ratio of option and stock price can be considered as a response/target variable in a learning approach whereas another observed scale-free quantity, the array of order statistics of mean-adjusted historical log returns, can be taken as a predictor/feature variable. By doing so, from the input asset price data, the magnitude, temporal, and drift information are removed and the spatial distribution of return is retained. On the other hand from the predicted target value, which stands for the ratio of option and spot prices, the predicted option price is obtained by multiplying the target with the observed stock price. Such an approach has great flexibility. It can learn from one asset class data and be applied to another, provided their log returns have similar distributions. See Goswami *et al.* (2021) for more details. We call such approaches as the Homogeneity Hint Approach or $\mathcal{A}_{HH}$.

However, requiring identical distributions of log returns in two different assets is quite restrictive, even under their risk-neutral measures. Unless the features and targets are further scaled, to factor out the large disparity of their distributions, the trained model fails to perform across assets. In other words, $\mathcal{A}_{HH}$ adapts poorly to domain shifts. Indeed such an approach may find feature values of the test data from another asset unfamiliar with reference to the training data and hence fail to perform for test dataset. In view of this, we wish to formulate a common representation space, i.e. a bridge between two different risk-neutral return distributions and utilize that

for connecting their corresponding option prices. To this end, we adopt parametric models of asset price dynamics.

2.2. Parametric settings

In this section and onwards we only consider the processes having continuous paths, unlike rcll process in the preceding section.

The simplest parametric model for the continuous-time dynamics of asset prices is geometric Brownian motion (GBM). The GBM which is also a Markov process satisfies (M1)–(M3), with no need of auxiliary process V and hence in particular (1) holds. Therefore, under GBM the theoretical call option price function $C(t, s; K, T; r, \sigma)$ satisfies

$$C(t, s; K, T; r, \sigma) = sC\left(t, 1; \frac{K}{s}, T; r, \sigma\right)$$

as a consequence of (A1). The function C is also known as the Black-Scholes-Merton (BSM) formula for a European call option. Here we replace $(S(t), V(t))$ by only $S(t)$, since S itself is Markov. As a result, the additional variable V is dropped from the option price function. The rationale behind such discrepancies in notation in other places is also evident from the context, so the discrepancies cause no ambiguity. Next, we consider an extension of BSM model for further discussion. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space, with a Brownian motion $W = W(t)_{t \geq 0}$ adapted to $\mathbb{F}$. We consider $S = \{S(t)\}_{t \geq 0}$ an Itô Process that models a stock, i.e. S satisfies the following SDE

$$dS(t) = \mu(t)S(t) dt + \sigma(t)S(t) dW(t), \quad (3)$$

where $\{\mu(t)\}_{t \geq 0}$ and $\{\sigma(t)\}_{t \geq 0}$ are adapted and bounded processes.

DEFINITION 2.4 (ρ -scaling of a process) *Given a positive valued process $S = \{S(t)\}_{t \geq 0}$, define $A := \{A(t)\}_{t \geq 0}$ such that $A(t)^\rho = S(t)$ for all $t \geq 0$ and for some $\rho > 0$. We call the process A as ρ -scaling of process S .*

Clearly, the ρ -scaling is an easy recipe for obtaining a parametric family of processes, all having different risk-neutral return distributions. A more precise version of this statement is given in the following lemma whose proof can be found in the appendix.

LEMMA 2.5 *Assume that S satisfies (3) and $\rho > 0$. (i) Both S and its ρ -scaling satisfy (M1)–(M3). (ii) The instantaneous volatility of the ρ -scaling of S is identical to the instantaneous volatility of S divided by ρ .*

Lemma 2.5 implies that although S and A satisfy (M1)–(M3), Theorem 2.2 is not applicable between them. In other words, an identity like (2) between the option prices on S and A does not hold, as their returns' distributions under the respective risk-neutral measures are not identical. However, given a pair of assets whose prices are modeled by (3) with different parameter values, one can choose a pair of ρ values so that the assets' ρ -scalings satisfy an identity like (2). This fact is stated in the following theorem. A proof of this is deferred to the appendix.

THEOREM 2.6 *Assume that for each $i = 1, 2$, S_i solves (3) with $\mu = \mu_i$, $\sigma = \sigma_i$ and Brownian motion $W = W_i$. For a fixed $t \in [0, T)$, denote*

$$\rho_i := \left(\frac{1}{T-t} \mathbb{E} \int_t^T \sigma_i^2(u) du \right)^{\frac{1}{2}}, \quad \forall i = 1, 2, \quad (4)$$

where $\mathbb{E}$ is the expectation with respect to $\mathbb{P}$, and denote the price function of a European call option on the ρ_i -scaling of i th asset, by ψ_i . Then

- (i) the risk-neutral distribution of log return of the ρ_i -scaling of i th assets are identical,
- (ii) for all $a_1 > 0$, $a_2 > 0$, $p > 0$

$$\frac{1}{a_1} \psi_1(t, a_1, pa_1, T) = \frac{1}{a_2} \psi_2(t, a_2, pa_2, T). \quad (5)$$

REMARK 2.7 Although ρ in (4) depends on the law of σ , and values of t , and T , we don't write it here to avoid clumsy notation. However, we refer to the value in (4) as *volatility scalar* to avoid ambiguity. For the rest of the paper, ρ and ρ_i denote the volatility scalars of S and S_i respectively, unless otherwise mentioned. A brief discussion on the estimation of the volatility scalar ρ is presented in the appendix.

In the aim of building a market-resilient learning approach for option price prediction, we follow the approach of common representation. First of all, we are required to select features that can help to extrapolate for successfully predict option prices corresponding to atypical out-of-sample data. In this connection, Theorem 2.6(i) is useful. In particular, it asserts that the risk-neutral distribution of the log return of the ρ -scaling of the asset price is a market-resilient feature influencing the option pricing mechanism. Needless to say, the supervised learning approach should also have features coming from contract parameters and risk-free assets. This feature representation from samples in different domains (training set and atypical test set) is indistinguishable.

Next, we are required to decide on a suitable target variable related to option price that depends solely on the market resilient features and not on factors excluded from the feature, for formulating the common representation space. There is no unique or obvious way to do that. For example, a relation between φ_i and ψ_i , the observable and unobserved option prices of the assets with prices S_i and A_i for each i , may be derived. Then a relation between φ_1 and φ_2 , the observable option prices of the assets with prices S_1 and S_2 can be derived, as Theorem 2.6 relates ψ_1 and ψ_2 . It seems that this is hard in general. If for each $i = 1, 2$,

$$\psi_i(t, a_i, pa_i, T) = F(\varphi_i(t, a_i^{\rho_i}, pa_i^{\rho_i}, T), a_i, \rho_i, p, T) \quad (6)$$

for some known F , then from Theorem 2.6 we have

$$\begin{aligned} & \frac{1}{s_1^{1/\rho_1}} F(\varphi_1(t, s_1, ps_1, T), s_1^{1/\rho_1}, \rho_1, p, T) \\ &= \frac{1}{s_2^{1/\rho_2}} F(\varphi_2(t, s_2, ps_2, T), s_2^{1/\rho_2}, \rho_2, p, T) = \mathcal{U} \text{ (say)} \end{aligned} \quad (7)$$

provided ρ_1 and ρ_2 are chosen as in (4). The target $\mathcal{U}$ is invariant for two different contracts with identical moneyness and

time to maturity on two different assets, possibly with domain shifts. Also, $\mathcal{U} = \psi_i(t, a_i, p a_i, T)/a_i$, which depends solely on the A_i features and the features related to the contract and risk-free asset. One may perhaps obtain some other targets based on a transformation involving implied volatility etc so that similar common representation is created. Assume that a learning approach $\mathcal{A}$ is trained to take centered return distribution of S_i^{1/ρ_i} (i.e. the returns of A_i after being subtracted by its mean) and variables and parameters p, T, r as inputs and to give $\mathcal{U}$ as target. Then $\mathcal{A}$ learns a fundamental property of option pricing from the common representation space, that holds across assets having a wide range of return distributions. Further, if the map $\varphi \mapsto \mathcal{U}$ is invertible, then for test data, the predicted target may be used for computing the option price. We explain this idea by making use of an approximation formula (Bharadia *et al.* 1996) in the next section.

3. Application of an implied volatility estimate in domain adaptation

3.1. A common representation space

In this section, instead of an equality (6), which is hard to find, we use an approximate equality relation to derive an approximation of the equality (7). To this end, we refer to Bharadia *et al.* (1996), which gives a simple approximation of implied volatility, the inversion of the BSM formula for near ATM options. The literature on the inversion of the BSM formula is reasonably rich. We cite Li (2005) for various other improvements over the simple formula in Bharadia *et al.* (1996) which we use in this section. For each asset $i = 1, 2$, the approximation formula for IV^{S_i, p_i, T_i} , the implied volatility of asset S_i corresponding to a call contract with moneyness and time to maturity p_i , and T_i respectively, is given by

$$IV^{S_i, p_i, T_i} \approx \sqrt{\frac{2\pi}{T_i} \left(\frac{\varphi_i(0, s_i, p_i s_i, T_i)}{s_i(1 + p_i^*)/2} - \frac{(1 - p_i^*)}{(1 + p_i^*)} \right)}$$

where $p_i \approx 1$, $p_i^* = p_i e^{-rT_i}$, s_i is the present price of i th stock. We recall that φ_1 and φ_2 are price functions of European call contracts on assets having prices S_1 and S_2 respectively which follow diffusion equations as in (3). Since, ρ_i is defined (see the definition (4)) using the instantaneous volatility values on the time interval between present and maturity time, the implied volatilities should nearly be proportional to ρ_i , provided all other parameters are kept constants. That is if $p_1 \approx p_2$, and $T_1 \approx T_2$, we have

$$\frac{IV^{S_1, p_1, T_1}}{\rho_1} \approx \frac{IV^{S_2, p_2, T_2}}{\rho_2}. \quad (8)$$

However, the stylized facts and other theoretical validations of volatility smile indicate that the said approximation should worsen as T_1 , and p_1 depart from T_2 , and p_2 . Finally, by plugging in the approximate expressions in the above approximate

equality we get

$$\begin{aligned} & \sqrt{\frac{2\pi}{T_1} \left(\frac{\varphi_1(0, s_1, p_1 s_1, T_1)}{s_1(1 + p_1^*)/2} - \frac{(1 - p_1^*)}{(1 + p_1^*)} \right)} / \rho_1 \\ & \approx \sqrt{\frac{2\pi}{T_2} \left(\frac{\varphi_2(0, s_2, p_2 s_2, T_2)}{s_2(1 + p_2^*)/2} - \frac{(1 - p_2^*)}{(1 + p_2^*)} \right)} / \rho_2, \\ & \text{or, } \mathcal{U}(s_1, r, \sigma_1, p_1, T_1, \rho_1) \\ & = \mathcal{U}(s_2, r, \sigma_2, p_2, T_2, \rho_2) \end{aligned} \quad (9)$$

where for $i = 1, 2$

$$\begin{aligned} & \mathcal{U}(s_i, r, \sigma_i, p_i, T_i, \rho_i) \\ & = \sqrt{\frac{2\pi}{T_i} \left(\frac{\varphi_i(0, s_i, p_i s_i, T_i)}{s_i(1 + p_i^*)/2} - \frac{(1 - p_i^*)}{(1 + p_i^*)} \right)} / \rho_i. \end{aligned} \quad (10)$$

We recall that for each $i = 1, 2$, σ_i/ρ_i is the volatility of A_i as a consequence of Lemma 2.5(ii). Again, (4) implies $(\frac{1}{T-i} \mathbb{E} \int_t^T (\sigma_i(u)/\rho_i)^2 du)^{\frac{1}{2}} = 1$. Hence, the volatility scalar value corresponding to each A_i is 1. Therefore, from (8), for each $i = 1, 2$,

$$\mathcal{U}(s_i, r, \sigma_i, p_i, T_i, \rho_i) \approx \frac{IV^{S_i, p_i, T_i}}{\rho_i} \approx \frac{IV^{A_i, p_i, T_i}}{1},$$

the implied volatility of asset A_i corresponding to a contract on it with identical moneyness and time to maturity. Hence, $\mathcal{U}(s_i, r, \sigma_i, p_i, T_i, \rho_i)$ may be regarded as primarily dependent on the features of asset price A_i and the features related to contract and risk-free asset. So, $\mathcal{U}$, as in (10) is a legitimate target variable for a common representation space, provided the approximation errors are considerably small.

3.2. Discussion on the approximation error

Equation (9) expresses an approximate relation between these two option prices and conveys that $\mathcal{U}$ is nearly constant wrt s, σ , and the volatility scalar ρ . In particular, if prices of a pair of stocks satisfy (3) with $\sigma_1 \stackrel{d}{=} \sigma_2$, then (A5) holds with $\rho_1 = \rho_2 = \rho$, for any arbitrary value. With $\rho_1 = \rho_2 = 1$, clearly $S_i = A_i$ and $\psi_i = \varphi_i$ for each $i = 1, 2$. Consequently (5) gives $\frac{\varphi_1(0, s_1, p_1 s_1, T)}{s_1} = \frac{\varphi_2(0, s_2, p_2 s_2, T)}{s_2}$. On the other hand if $T_1 = T_2 = T$, $p_1 = p_2 = p$, equation (9) gives

$$\begin{aligned} & \left(\frac{\varphi_1(0, s_1, p s_1, T)}{s_1} - \frac{(1 - p^*)}{2} \right) / \rho \\ & \approx \left(\frac{\varphi_2(0, s_2, p s_2, T)}{s_2} - \frac{(1 - p^*)}{2} \right) / \rho \\ & \text{i.e. } \frac{\varphi_1(0, s_1, p s_1, T)}{s_1} \approx \frac{\varphi_2(0, s_2, p s_2, T)}{s_2}. \end{aligned}$$

Thus the relation in (9) is exact, i.e. ‘ $\approx$ ’ can be replaced by ‘ $=$ ’, when the volatility scalars are identical, i.e. $\rho_1 = \rho_2$ and the contract moneyness and time to maturities match.

The approximation error in (9) associated to a pair of similar contracts (i.e. $T_1 = T_2 = T$, $p_1 = p_2 = p$) on assets

having volatility processes σ_1 and σ_2 is measured by the relative error $\sup\{\frac{|\mathcal{U}_1 - \mathcal{U}_2|}{\max(\mathcal{U}_1, \mathcal{U}_2)} \mid s_1 > 0, s_2 > 0\}$ where, $\mathcal{U}_i = \mathcal{U}(s_i, r, \sigma_i, p, T, \rho_i)$. We compute this error for the ATM options, i.e. moneyness parameter $p = 1$ under the special case of BSM models with constant volatility coefficients. The time to maturity T is taken as 0.2 year. In figure 1(a), the horizontal and vertical axes represent the volatility coefficients values of two different assets. Each volatility value ranges on $[5\%, 100\%]$. The darkness of shades show the relative error values for each ordered pair (σ_1, σ_2) . The maximum error is less than 4.5%. If none of σ_1 and σ_2 is smaller than 9%, the error is less than 2%. This numerically validates that (9) is a reasonable approximation for practical purposes where the volatilities are not too small.

It is evident from figure 1(a) that the error is minimal when σ_1 and σ_2 are identical. This can be explained by appealing to the homogeneity hint, stated under the non-parametric setting in Theorem 2.2. A separate visualization of error's dependence on the degree of mismatch of the volatilities is given in the scatter plot figure 1(b) under the same setting. In this $\frac{\max(\sigma_1, \sigma_2)}{\min(\sigma_1, \sigma_2)}$, a measure of mismatch, is plotted on the horizontal axis and the corresponding relative error of the approximation (9) is plotted on the vertical axis. Every blue circle corresponds to a pair of volatilities (σ_1, σ_2) for which the relative error is computed. The horizontal and the vertical coordinates of each circle are $\frac{\max(\sigma_1, \sigma_2)}{\min(\sigma_1, \sigma_2)}$ and $\sup\{\frac{|\mathcal{U}_1 - \mathcal{U}_2|}{\max(\mathcal{U}_1, \mathcal{U}_2)} \mid s_1 > 0, s_2 > 0\}$ respectively, where σ_i is the constant volatility of i th asset. It shows that if the largest volatility is not larger than double the smallest, the relative error is less than 1.5%. On the other hand, the error is less than 3% if the largest is not larger than 9 times the smallest.

3.3. Approach $\mathcal{A}_{DS}$ for domain shift

Assume that a learning approach takes risk-neutral return distribution of ρ_1 -scaling A_1 of asset prices S_1 as a feature variable. To be more precise, the order statistics of mean subtracted log returns of A_1 for the past few time periods are included in the feature set. The approach should also have features from contract parameters p and T , and risk-free assets' interest rate r . Also, set $\mathcal{U}(s_1, r, \sigma_1, p_1, T_1, \rho_1)$ as the target, which can be obtained from the observed option value φ_1 in the training data set. Consequently, for a test data set on another asset S_2 , after feeding in the feature values to the trained model, a value of the target $\mathcal{U}(s_2, r, \sigma_2, p_2, T_2, \rho_2)$ is predicted. Then the option price can be revived using the following formula

$$\begin{aligned} \varphi_2(t, s_2, p_2, T_2) \\ = s_2 \left(\rho_2 \frac{1 + p_2^*}{2} \sqrt{\frac{T_2}{2\pi}} \mathcal{U}(s_2, r, \sigma_2, p_2, T_2, \rho_2) + \frac{1 - p_2^*}{2} \right), \end{aligned} \quad (11)$$

irrespective of its volatility as long as ρ_2 is the volatility scalar for S_2 . Therefore, the above-mentioned learning approach is better adapted for the domain shift. We denote this as $\mathcal{A}_{DS}$. Of course, it is possible to develop a different learning model following this idea by choosing an approximation relation more accurate than (9) for possibly better performance. So, as far

$\mathcal{A}_{DS}$ is concerned, we can imagine that although this simple approximation facilitates predicting option prices reasonably well for test data sets with domain shifts, would fail to outperform $\mathcal{A}_{HH}$ when domain shift is absent. This raises the need of an ensemble model.

3.4. Ensemble model

The difference between the present and the past mean volatilities, normalized by the second one, may be considered as a crude measure of the domain shift. We call this measure the domain shift quotient or simply DSQ. To be more precise, if σ_0 and σ_i are the average historical volatility of the full training data set and the current historical volatility associated with i th test data respectively, the formula for the DSQ for the specific test data is $\frac{|\sigma_i - \sigma_0|}{\sigma_0}$. In this subsection, we propose an ensemble model where the ratio of weights given to the predictions from $\mathcal{A}_{DS}$ and $\mathcal{A}_{HH}$ models is $\lambda_1 (\frac{|\sigma_i - \sigma_0|}{\sigma_0})^{\lambda_2}$, a power function of DSQ. We denote the resulting approach as $\mathcal{A}_{E(\lambda)}$ where $\lambda = (\lambda_1, \lambda_2)$. If $P_{DS}(i)$ and $P_{HH}(i)$ are the predicted prices of i -th option from the models with approaches $\mathcal{A}_{DS}$ and $\mathcal{A}_{HH}$ respectively trained with same data, then $P_E(i)$, the predicted price from an ensemble approach $\mathcal{A}_{E(\lambda)}$, is defined as

$$\begin{aligned} P_{E(\lambda)}(i) = & \frac{1}{1 + \lambda_1 \left(\frac{|\sigma_i - \sigma_0|}{\sigma_0} \right)^{\lambda_2}} P_{HH}(i) \\ & + \frac{\lambda_1 \left(\frac{|\sigma_i - \sigma_0|}{\sigma_0} \right)^{\lambda_2}}{1 + \lambda_1 \left(\frac{|\sigma_i - \sigma_0|}{\sigma_0} \right)^{\lambda_2}} P_{DS}(i). \end{aligned} \quad (12)$$

Here λ_1 controls the scale of the adjustment and λ_2 controls the sensitivity of the response. As a result, for a fixed λ , the weight to P_{HH} diminishes as the atypicality rises in the test data. The more atypical the return data, i.e. the larger the relative difference between the test and the training volatilities, the higher weight is assigned to the prediction P_{DS} in the ensemble approach $\mathcal{A}_{E(\lambda)}$. The parameter λ is to be estimated using the least square method from a validation set. For effective tuning of λ , the validation set is required to have both typical and atypical data. The ensemble approach using the estimated λ may simply be denoted by $\mathcal{A}_E$.

4. Data

According to the Futures Industry Association (FIA), 84% of global equity options are now traded on Indian exchanges, up from just 15% a decade ago.[†] In the first quarter of 2024 alone, over 34.8 million equity index options were traded on India's two main financial derivatives exchanges-the National Stock Exchange of India (NSE) and the Bombay Stock Exchange (BSE) India. This was double the volume from Q1 2023 and accounted for 73% of all futures and options traded globally. Despite this high trading volume, the market's value

[†] Article by Bennett Voyles <https://www.fia.org/marketvoice/articles/premium-turnover-indian-options-hits-150-billion>

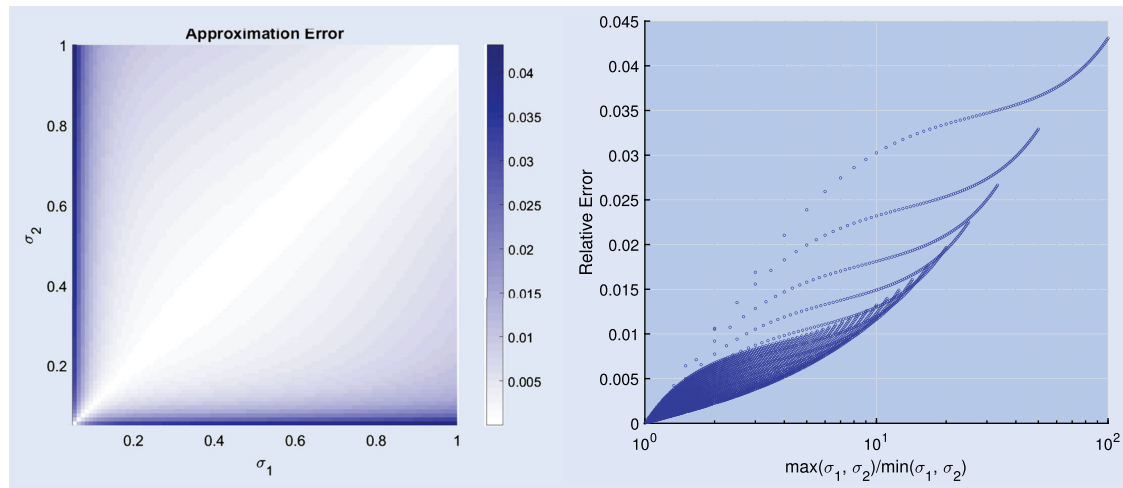


Figure 1. Visualization of relative error in the approximation relation (9). (a) Grid heatmap of the approximation error for a pair of volatilities and (b) The approximation error vs disparity(max-min ratio) of volatilities.

remains relatively small compared to North American and European markets due to lower notional values and premiums per contract. For example, in March 2024, NSE's total options premiums were \$150 billion, about a quarter of the \$598 billion seen across all US options exchanges. Nonetheless, the Indian market's rapid growth and substantial trading volume makes it a crucial area for developing and applying data-driven option trading strategies.

4.1. Description

For our study, we analyze daily contract price data for European call options on NIFTY50[†] and BANKNIFTY[‡]. The abundance of data allows a learning model to better capture and depict market behavior and trading patterns. On the other hand, due to the presence of many traders' participation, the traded prices are free from market inefficiency. The dataset, spanning January 1, 2015, to April 30, 2020, is obtained from the NSE's historical archives[§]. The data-driven supervised learning approaches for pricing European-style call options, outlined in Goswami *et al.* (2021) reportedly performed worse in abrupt highly volatile market conditions. Our objective is to introduce a novel market-resilient approach for improving the prediction for out-sample data having atypical asset returns. To this end, we incorporate testing of the trained model with data from the COVID-19 lockdown period, when the financial market crashed, i.e. from January 2020 to April 2020.

Each data point includes the following information, (i) the daily closing prices of the underlying asset, (ii) moneyness and the time to maturity of a particular European call option, (iii) daily 19 log returns of the underlying asset from the 20 most recent daily data, (iv) daily close price of the same call

contract on the current and earlier day and (v) prevailing risk-free interest rate. Moreover, we filter out all data points for which the same option contract did not exist the previous day. Our choice of a 20-day window aligns approximately with a calendar month, accounting for holidays. Since we compute the daily 19 log returns of the underlying asset, to have these log returns from January 1, 2015, to April 30, 2020, we consider the daily data of the underlying assets (specifically, NIFTY50 and BANKNIFTY indices) from October 1, 2014, to April 30, 2020.

We retain datapoints only for option contracts that are near at-the-money (ATM), specifically those where the ratio of the spot price to the strike price is between 96% and 104%. We refer to these as near-ATM option contracts. It is observed that numerous near-ATM contracts are traded daily, with varying times to maturity. Since, contracts with large or minimal times to maturity have significantly low trading volumes, we focus on studying contracts with time-to-maturity values between 3 and 45 days.

4.2. Train/Test split:

To develop a predictive model using XGBoost supervised learning, we partitioned the dataset into three segments, allocating roughly 70% for training, 10% for validating, and 20% for testing the trained models. The split is such that the training set includes data only dated before the oldest data of the validation set and the latest data of the validation set is older than all data in the testing set. This is for avoiding information leakage as explained in Wang and Ruf (2022). Specifically, the training and validation datasets span 52 and 4 months, covering the period from 1st January 2015 to 30th April 2019, and 1st May 2019 to August 31, 2019 respectively.

Subsequently, the test dataset is subdivided into two parts to facilitate comprehensive evaluation:

- (1) **Typical:** The 'typical' test dataset encompasses data from September 01, 2019, to December 31, 2019. This period is characterized by typical market dynamics, enabling model evaluation under market behavior similar to that observed during the training phase.

[†] NIFTY 50, abbreviated as NIFTY50, serves as a benchmark index for the Indian stock market. It reflects the weighted average performance of 50 of the largest companies listed on the National Stock Exchange of India.

[‡] The NIFTY Bank index, referred to as BANKNIFTY, consists of the most liquid and highly capitalized Indian banking stocks.

[§] Link to access the data -https://www1.nseindia.com/products/content/derivatives/equities/historical_fo.htm

Table 1. Train/Test split of European style call option contract dataset sizes.

Dataset	Time Period	N50	BNF
Raw	01/01/2015 – 30/04/2020	1 611 549	576 302
Filtered	01/01/2015 – 30/04/2020	27 372	37 284
Train	01/01/2015 – 30/04/2019	17 354	24 530
Validation	01/05/2019 – 31/08/2019	1828	1818
Typical Test	01/09/2019 – 31/12/2019	3692	3709
Atypical Test	01/01/2020 – 30/04/2020	3047	2774

- (2) **Atypical:** The ‘atypical’ test dataset consists of data from January 01, 2020, to April 30, 2020. This period includes the COVID-19 Indian financial market crash, presenting a unique testing scenario to assess model robustness under an extremely volatile market. The dynamics during this period are significantly different from those observed during the training phase, thereby offering valuable insights into the model’s performance in unprecedented market scenarios.

Table 1 provides a breakdown of the number of data points at each stage of the model building and evaluation process, offering insight into the dataset’s composition and distribution across training and test phases.

It is important to highlight that the sum of data points in the training, typical test, and atypical test datasets do not match the number of data points in the filtered dataset. This discrepancy occurs because, during the feature generation step in our approach, we remove the rows of data associated with option contracts that do not have a previous day option closing price, which is a feature in our learning approach. A brief exploratory data analysis is presented below. The Q-Q plot in figure 2 shows that the log returns of daily closing values of NIFTY50 and BANKNIFTY indices do not possess identical distribution. Nevertheless they are not significantly different during the period that constitutes the training dataset. The Q-Q plot is linear and has a slightly different slope than the identity line. Next, we compare the statistical behavior of log-returns of daily closing values of NIFTY50 index during different time intervals. In figure 3(a) the said comparison between the data from training window and the typical test window is presented. On the other hand, figure 3(b) presents a comparison between the data from training and the atypical test windows. In figure 3(a) the Q-Q plot is aligned with the identity line but in figure 3(b) the slope of the Q-Q plot is visibly far from the identity line. Similar comparisons are reported in figure 4 where we compare the distributions of log-returns of daily closing values of BANKNIFTY index during different time intervals. In figure 4(a), except few outliers, the return distribution of the typical test dataset matches well with that of the training dataset. The points in the Q-Q plot approximately lie on the identity line. However, when the atypical test data is compared with the training data in figure 4(b), the distributions of log-returns of the closing values appear to differ drastically. Although the Q-Q plot is linear in this case, the slope vastly mismatches with that of the identity line. These observations justify the use of terminology ‘typical’ and ‘atypical’ for two non-overlapping test datasets.

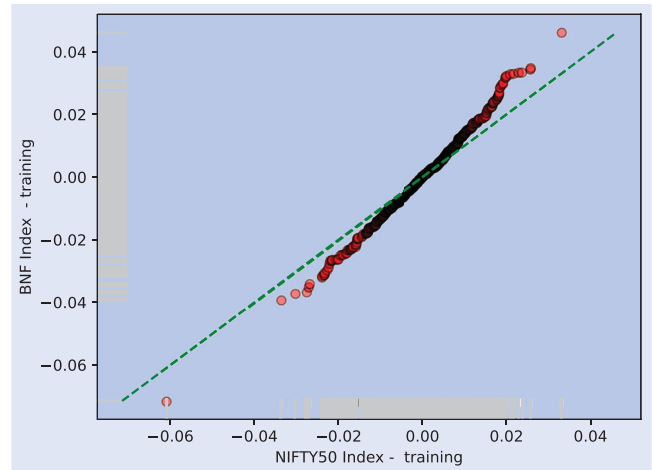


Figure 2. Q-Q plot for the log return distributions of the ‘Close’ prices of the BANKNIFTY and NIFTY50 indices for training dataset from January 01, 2015 to April 30, 2019.

5. Description of supervised learning approach

In this section, we outline the steps involved in the supervised learning approach utilized in this study. A schematic flowchart of the model pipeline is added in figure 5 for better clarity.

Step 1: Data Cleaning and Filtering In this essential step, we prepare the gathered raw data for European call option contracts to apply the ML algorithms. Initially, we eliminate rows containing blank ‘-’ entries in the ‘Underlying Value’ or ‘Strike Price’ columns. Additionally, any rows containing values of ‘0’ in the ‘Open’ or ‘Close’ columns are removed.

To further refine the selection, we follow the modularity approach of Goswami *et al.* (2021) and focus on option contracts that closely resemble at-the-money (ATM) contracts. This involves filtering contracts based on a criterion where the absolute quantity $|1 - \frac{S}{K}|$ is less than or equal to the pre-decided value of 0.04 as considered in Goswami *et al.* (2021), where S and K respectively represent the spot price of underlying and strike price of the option contract. However, it is important to note that contracts with significantly large or small time to maturities exhibit notably diminished trading volumes. Therefore, to conduct a more focused analysis, we opt to exclusively investigate contracts with time-to-maturity (‘TTM’) values falling within the range of not more than 45 days and not less than 3 days.

Step 2: Feature Generation Let Δ denote the time granularity in year unit and $\tilde{S}(t) := S(\Delta * t)$ the price of the risky asset at present time $\Delta * t$, where t is an integer. The discrete time series $\{\tilde{S}(t) \mid t = 0, 1, \dots\}$ gives daily data if $\Delta = 1/255$, by imagining 255 number of trading days in a year. Let n be a fixed natural number. For each $i \leq n$, the feature variable $F_i := R_{(i)}$, where, $\{R_{(i)} : i = 1, \dots, n\}$ is order statistics of $R_i := \ln(\tilde{S}(t - n + i)/\tilde{S}(t - n - 1 + i)) - \frac{1}{n} \ln(\tilde{S}(t)/\tilde{S}(t - n))$, the centered daily log return. The order statistics $R_{(i)}$ are determined by arranging R_i in ascending order for each sample. To clarify, the i^{th} order statistic, denoted by $x_{(i)}$, of a sample comprising various real values $(x_1, x_2, x_3, \dots, x_n)$ represents the i^{th} -smallest value, such that $x_{(i)} = x_{\beta(i)}$ where β is a permutation on $\{1, \dots, n\}$, and $x_{(1)} \leq$

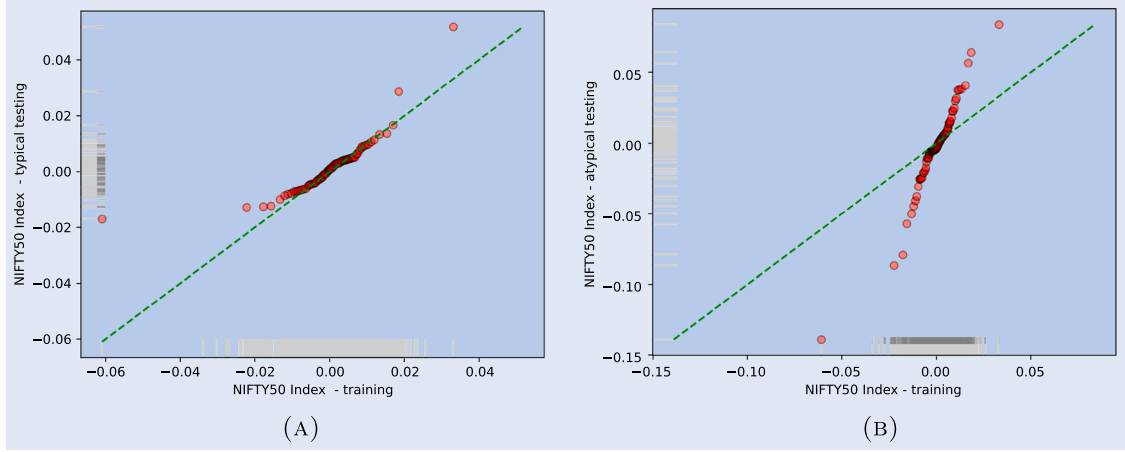


Figure 3. Q-Q plot for the log return distributions of the ‘Close’ prices of NIFTY50 in training and testing data. (a) Train vs Typical test data and (b) Train vs Atypical test data.

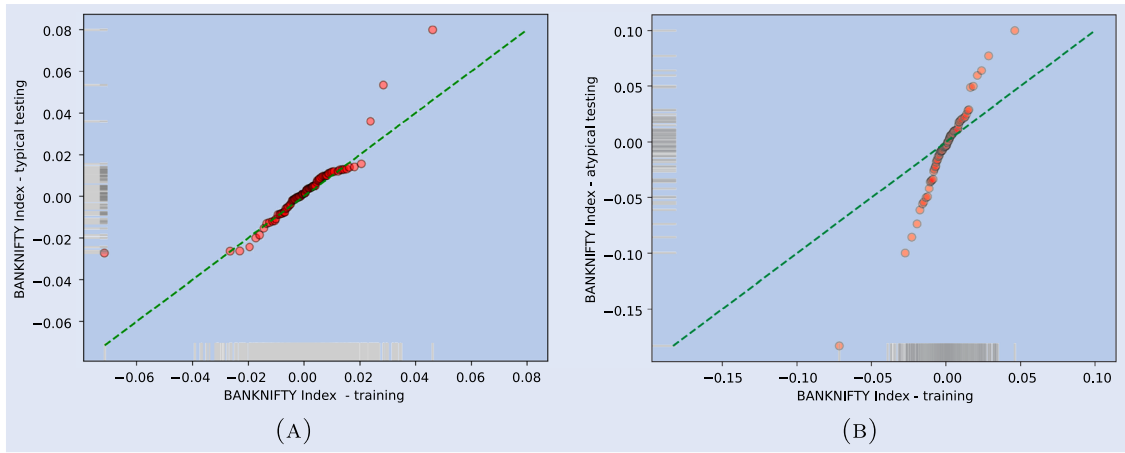


Figure 4. Q-Q plot for the log return distributions of the ‘Close’ prices of BANKNIFTY in training and testing data. (a) Train vs Typical test data and (b) Train vs Atypical test data.

$x_{(2)} \leq \dots \leq x_{(n)}$ is satisfied. We set $n = 19$ for sampling past daily data of nearly one month. At this stage, we compile a set of 24 features to be utilized in supervised learning models:

- The time-to-maturity (‘TTM’) of the option contract under study, measured in days.
- The moneyness of the option contracts is taken as a feature. It is calculated as the ratio $\frac{K}{S}$, representing the ratio between the strike price K of the option contract and the underlying value or Spot Price S being examined.
- The ideal bank’s interest rate is approximated by the three-month sovereign bond yield rates in the literature and added as a feature. This serves as an estimation for risk-free interest rates.
- For time index $t \geq 19$, 19 order statistics of centered log return, denoted by F_1 through F_{19} , derived from index (or underlying asset) data make up 19 feature variables for the $\mathcal{A}_{HH}$ model. That is $F_i = R_{(i)}$ for all $i \leq 19$. Also we define another feature variable $F_{20} = \frac{1}{19} \ln(\tilde{S}(t)/\tilde{S}(t-19))$. On the other hand, for the $\mathcal{A}_{DS}$ model, we divide each of these statistics by ρ , the volatility scalar, to get its 20 feature variables. The volatility scalar ρ is estimated

using historical volatility, the square root of annualized (assuming 255 trading days in a year) variance computed from the set of feature variables F_1 to F_{19} , derived earlier.

- The observed target variable value at the previous time is considered as a feature too. The formula for this feature, of course, depends on the approach. If $\text{Trgt}(t)$ denotes the observed value of target at time index t , then $\text{Trgt}(t-1)$, the value at time index $t-1$, is known at time index t , and is used as a feature at present, having time index t . The target in both approaches depend on close price of the option contract. So, $\text{Trgt}(t-1)$ depends on $C(t-1)$. If the option contract was not traded on the previous day (i.e. $C(t-1)$ data is unavailable), we exclude this option from consideration.

Step 3: Target Variable We use regression to implement the Homogeneity Hint Approach $\mathcal{A}_{HH}$, as described in Remark 2.3. According to this approach (also see Goswami *et al.* 2021) ‘ $\mathcal{A}_{HH}$ target’ is the ‘normalized price’ and is given by

$$\mathcal{A}_{HH} \text{ target}(t) = 100 \times \frac{C(t)}{\tilde{S}(t)}, \quad (13)$$

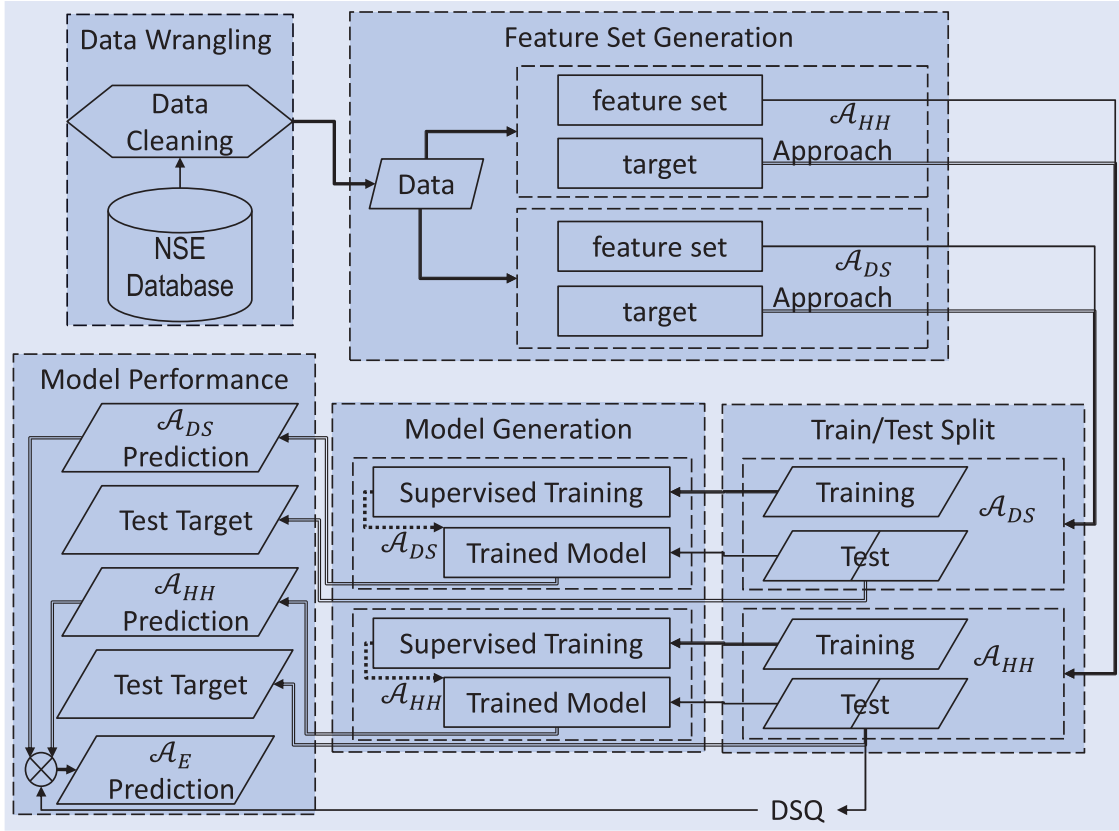


Figure 5. A schematic flowchart of the model pipeline.

a scale-free target variable. For the regression implementation of the $\mathcal{A}_{DS}$ approach, as presented in Subsection 3.3, the scale-free target variable referred to as the ' $\mathcal{A}_{DS}$ target' is defined by (9), i.e.

$$\mathcal{A}_{DS} \text{ target}(t) = \frac{1}{\rho} \sqrt{\frac{2\pi}{T}} \left(\frac{C(t)}{\bar{S}(t)(1+p^*)/2} - \frac{(1-p^*)}{(1+p^*)} \right).$$

where C , S , and T denote the call option price, underlying value, and time-to-maturity in years, respectively. Here, $p = \frac{K}{S(t)}$ is moneyness, $p^* = \frac{Ke^{-rT}}{S(t)} = pe^{-rT}$ is discounted moneyness, where K is the strike price of the call option contract and r is the interest rate of an ideal bank estimated to 'Yield03' value divided by 100.

Step 4: Ensemble Model At this step we implement the ensemble approach (12) formulated in Subsection 3.4. Since the total availability of atypical data is limited, for statistically meaningful performance analysis, all atypical data is kept for testing. Consequently, the validation set is devoid of any atypical data. However, for effective tuning of λ , the validation set is required to have both typical and atypical data. Anyways, in view of the primary goal of the investigation, we give up tuning the parameters λ_1 and λ_2 , and set a canonical value $\lambda_1 = \lambda_2 = 1$. That is the ratio of weights given to the predictions from $\mathcal{A}_{DS}$ and $\mathcal{A}_{HH}$ is identical to DSQ. As a result, we have

$$P_{E(1,1)} = \frac{1}{1 + DSQ} P_{HH} + \frac{DSQ}{1 + DSQ} P_{DS}.$$

Although due to this simplicity, the performance may not be superior, however, it is enough for us to demonstrate the

Table 2. Hyperparameter values for XGBoost.

Name	Value Set
n_jobs	4
objective	reg:squarederror
n_estimators	500
max_depth	6
learning_rate	0.01
min_child_weight	3
colsample_bytree	1.0
subsample	0.6

promise of an ensemble model, which is the primary goal of this study.

Step 5: Training and Setting Numerical Values of Hyperparameters We ran the codes in Python version 3.10.4. XGBoost, a well-studied supervised ML algorithm developed by Chen and Guestrin (2016), was utilized in this research. Table 2 presents the hyperparameter values used to train the XGBoost (Extreme Gradient Boosting) linear regression model described in this manuscript. While XGBoost offers numerous hyperparameters, we focused on the following key ones. For detailed information on each hyperparameter used, please refer to the XGBoost Documentation[†].

To balance the bias and variance, we use validation set for tuning the above-mentioned hyperparameters. We choose the values that optimize the performance of trained models on the

[†] There are many resources available to read about XGBoost parameters; for example, one can refer to <https://xgboost.readthedocs.io/en/stable/parameter.html>

validation set, and used those values for all subsequent numerical experiments. This allowed the models to gain better generalizability. We conduct a cross-validated grid search using the scikit-learn GridSearchCV routine to determine the optimal values for these hyperparameters. However, we observed that the model performance did not vary significantly with changes in the hyperparameters.

Step 6: Testing and Error Metric After the training phase, it is imperative to evaluate our model's performance using unseen test data, ensuring a comprehensive understanding of prediction quality. This study employs Root Mean Square Error (RMSE) to assess regression model predictions. Assume that the total number of data points in the test set is N and e_i is the difference between the true value of (13) and its predicted value, coming from the target of a model $\mathcal{M}$ at the i th entry. Then e_i represents the prediction error for the i th test data for each $i \leq N$. Finally, the RMSE of the prediction of model $\mathcal{M}$ for the test data set is defined as

$$\text{RMSE} = \left(\frac{1}{N} \sum_{i=1}^N (e_i)^2 \right)^{\frac{1}{2}}. \quad (14)$$

It is crucial to emphasize that the evaluation specifically centers on gauging the regression model's performance with respect to the scale-free target variable denoted as 'normalized price' (13). In the context of ' $\mathcal{A}_{HH}$ target' scenarios, RMSE formula are applied directly to its target variable 'normalized price'. However, for scenarios involving ' $\mathcal{A}_{DS}$ target', we utilize (11) to compute the predicted 'normalized price' and subsequently evaluate the regression model's performance.

6. Model performance

Our study involves analyzing and comparing the performance of four approaches, namely, $Go(NN)$ (an ANN model adopted from Goswami *et al.* 2021), $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$, and $\mathcal{A}_{E(1,1)}$ for option price predictions. While the first three are supervised learning models, the last one is an ensemble model. The first and second approach $Go(NN)$ and $\mathcal{A}_{HH}$ are adopted from Goswami *et al.* (2021), whereas the other two are proposed in this article. We have collected historical option price data on two different indices N50 and BNF. Based on the data from each symbol we train four models using above mentioned four approaches. Moreover, we perform multi-source training using historical data from both symbols to get an additional combined model using each of the four approaches. Hence we obtain a total of 12 models. We test each of these 12 models on all four disjoint test data samples. It is worth noting that $Go(NN)$ and $\mathcal{A}_{HH}$ have identical features and targets. The former is an ANN model and the latter is an XGBoost model. For training $Go(NN)$, we have used the hyperparameter values as reported in Goswami *et al.* (2021). Since, prediction from $Go(NN)$ has no effect on prediction of $\mathcal{A}_{E(1,1)}$, the principal model in this study, we will exclude a detailed comparative discussion on $Go(NN)$ in the following subsections. We mention its performance as we treat $Go(NN)$ as a data-driven benchmark model.

Table 3. Performance for $Go(NN)$, $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$, and $\mathcal{A}_{E(1,1)}$ Approaches are assessed using RMSE on four disjoint test datasets.

RMSEs of option price prediction models					
Symbol(s) of Training data	Approach	Test - N50		Test - BNF	
		Typical	Atypical	Typical	Atypical
Benchmark - BSM (HV) N50	$Go(NN)$	0.678	1.679	0.966	1.533
	$\mathcal{A}_{HH}$	0.348	1.054	0.610	0.911
	$\mathcal{A}_{DS}$	0.222	1.303	0.405	1.278
	$\mathcal{A}_{DS}$	0.338	0.574	0.461	0.539
	$\mathcal{A}_{E(1,1)}$	0.205	0.484*	0.289	0.456*
BNF	$Go(NN)$	0.310	0.891	0.436	0.894
	$\mathcal{A}_{HH}$	0.214	1.121	0.256	1.103
	$\mathcal{A}_{DS}$	0.301	0.822	0.366	0.720
	$\mathcal{A}_{DS}$	0.211	0.669	0.232	0.574
	$\mathcal{A}_{E(1,1)}$	0.211	0.669	0.232	0.574
N50+BNF	$Go(NN)$	0.299	0.947	0.456	0.917
	$\mathcal{A}_{HH}$	0.217	1.105	0.263	1.097
	$\mathcal{A}_{DS}$	0.268	0.689	0.334	0.610
	$\mathcal{A}_{DS}$	0.198*	0.586	0.228*	0.507
	$\mathcal{A}_{E(1,1)}$	0.198*	0.586	0.228*	0.507

6.1. Performance comparison using RMSE

The model performances are presented in table 3. In that, the models are classified into 3 classes based on the training data sets. The symbol/symbols of each training data are listed in the first column. The approaches used in the model are mentioned in the second column. The option price prediction using historical volatility and the Black-Scholes-Merton formula is set as the benchmark for comparison. It is important to note that neither $\mathcal{A}_{HH}$ nor $\mathcal{A}_{DS}$ uses the IV in the feature variables or the target variable. Nevertheless, we have brought in the notion of IV in Subsection 3.1 just to explain theoretically how the proposed target in $\mathcal{A}_{DS}$ meets the goal of domain adaptation. In summary, since none of the approaches use IV explicitly, it is not unfair to avoid IV in computing the benchmark for a comparison. However, since IV is known to hold more information about option prices, we present table A2 separately in Appendix 2 to illustrate the performance of IV-based predictive models.

In table 3, we compare the performance of all these models and the benchmark using two types of test data coming from each of the two symbols N50 and BNF. The performance, i.e. the RMSE of predictions, of each model is reported for all four disjoint test data samples. Given a training symbol and test data, the best-performing approach is indicated by highlighting the lowest RMSE value in orange. Given a test dataset, the lowest RMSE across all 12 trained models is indicated with a star (*) sign.

At the outset, we notice that for each test data, the column-wise reported RMSE values for all nine training models based on $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$ and $\mathcal{A}_{E(1,1)}$ are smaller than the RMSE of the benchmark prediction. In fact, the lowest RMSE turns out to be smaller than 35% of the benchmark RMSE for each of the four test data sets. Moreover, given every pair of training symbol and test dataset, the best-performing approach is $\mathcal{A}_{E(1,1)}$. This empirically confirms the suitability of the ensemble approach for predicting option prices. It is further observed that on every atypical dataset, the $\mathcal{A}_{DS}$ approach outperforms $\mathcal{A}_{HH}$ approach, no matter what the training data set is. This

validates the usefulness of the domain shift approach presented in this paper. On the other hand, we also observe that $\mathcal{A}_{HH}$ outperforms $\mathcal{A}_{DS}$ on the typical test data of each symbol when the training data is from the same symbol. This empirically validates the theoretical conclusion of Theorem 2.2. We explain this below. We recall that $\mathcal{A}_{HH}$ is founded on perfect equalities, whereas $\mathcal{A}_{DS}$ is based on some approximations. So, in the special circumstances where Theorem 2.2 is applicable, $\mathcal{A}_{HH}$ gives a better result than $\mathcal{A}_{DS}$. On the other hand, for typical test data of the same symbol of training data, as anticipated, the assumption of similarity between return distributions of training and test is valid. Again, for the same reason, in the multi-source training scenario (see the row with heading N50+BNF), $\mathcal{A}_{HH}$ outperforms $\mathcal{A}_{DS}$ on both of the typical data. This justifies the benefit of combined training. Finally, we recall that the Q-Q plot in figure 2 exhibits some degree of similarity between N50 and BNF log returns. This explains why, for each typical test dataset, $\mathcal{A}_{HH}$ outperforms $\mathcal{A}_{DS}$ even when the training symbol is not the same as the test symbol.

It is interesting to note that each of the 9 models performs better on typical test data of N50 than those of BNF. On the contrary, each of the 9 models performs better on atypical test data of BNF than those of N50. Such consistent performance of approaches on typical or atypical test data irrespective of the source of training data indicates the importance of consideration of domain shifts in the model building. It also indicates that while $\mathcal{A}_{HH}$ is insensitive to the domain shift, $\mathcal{A}_{DS}$ incorporates the domain adaptability reasonably well. So far prediction on BNF atypical test data is concerned, RMSE value of the ensemble model trained with N50 data is lower than that of the ensemble model trained with BNF data. This is another instance of the success of common representation space and domain adaptation.

6.2. Performance of multi-source training model

Now we contrast the performance of multi-source or combined training with single-source training models. We start with the following simple observation from table 3. We consider the performance of models trained on data of one symbol and tested on typical/atypical data of another symbol with the approach $\mathcal{A}_{HH}$ or $\mathcal{A}_{DS}$. RMSEs of eight such cross-symbol experiments are reported in table 3. We compare those with eight other corresponding RMSEs for the multi-source models. In six out of eight cases, the RMSE of multi-source training is lower than that of cross-symbol experiments. For N50 typical test data, the RMSE of the multi-source training $\mathcal{A}_{HH}$ model is larger by 1.4% than that of the BNF-trained $\mathcal{A}_{HH}$ model. We also observe the following. Although $\mathcal{A}_{HH}$ approach performs poorer than $\mathcal{A}_{DS}$ on atypical test data, the performance of $\mathcal{A}_{HH}$ approach on atypical test data gets better in combined training model, compared to single-source training model. A similar statement also holds for the performance of $\mathcal{A}_{DS}$ models on typical data. These again illustrate the benefit of combined training. Finally, for every typical and atypical test data, the performance of the multi-source ensemble model is the best, and close to the best, respectively. To be more precise, the RMSE value for the multi-source ensemble model is

less than 35% of the RMSE of the benchmark on each of the four test datasets. This is empirical evidence of the reliability of the multi-source ensemble model.

The option price prediction performance of multi-source models with approaches $\mathcal{A}_{HH}$ and $\mathcal{A}_{DS}$, summarized in table 3, is illustrated more effectively through a histogram of residuals, defined as the differences between predicted and actual normalized price of options, as shown in figure 6. The horizontal axis represents the range of residuals, which is divided into 100 intervals, while the vertical axis shows the relative frequency. The histograms in figure 6 is constructed using predictions from the XGBoost models, trained on combined datasets, and tested on NIFTY50 (plot 6(a)) and BANKNIFTY (plot 6(b)) datasets respectively. These predictions are made using both the $\mathcal{A}_{DS}$ and $\mathcal{A}_{HH}$ approaches, under typical and atypical testing scenarios. Hence each plot has four histograms.

The performance reported in table 3 indicates that under typical testing scenarios, the $\mathcal{A}_{HH}$ approach yields better accuracy than $\mathcal{A}_{DS}$ approach, as corroborated by the histogram. In both subplots, the residuals from $\mathcal{A}_{HH}$ approach (represented by the ‘blue’ bars) are more tightly clustered around zero compared to the $\mathcal{A}_{DS}$ approach (‘orange’ bars). However, the difference in frequency levels between the two approaches is less pronounced, indicating the absence of drastic difference in error margin. On the N50 as well as BNF typical test data, residual distributions for both the approaches exhibit symmetric and low dispersion around zero. This confirms the absence of bias in the prediction.

The dispersion of prediction residual for atypical test data appears to be significantly larger than that of the typical data scenario for every approach in both figure 6(a,b). Additionally, an intriguing observation in the atypical testing scenario is that the $\mathcal{A}_{DS}$ approach leads to some outliers of overestimation of option prices relative to actual values, as evidenced by the leftward skew of the ‘red bars’, while the $\mathcal{A}_{HH}$ approach leads to outliers of underestimation of prices, as indicated by the rightward skew of the ‘green’ bars across both atypical testing indices. Hence the ensemble model performs significantly better on atypical data by neutralizing such polar outliers in $\mathcal{A}_{HH}$ and $\mathcal{A}_{DS}$ predictions. In summary, the histograms provide a clearer and deeper insight into the models’ performance across different testing scenarios than the RMSE errors offer in table 3.

6.3. Model performance on synthetic test data

Now we present the results of two experiments conducted to assess the performance of trained models on synthetic data. The use of synthetic data overcomes real-world data limitations for assessing domain adaptation in rare but large domain shifts. The synthetic data is simulated using Black-Scholes-Merton model of asset prices and the theoretical fair prices of options. The primary goal of these experiments is to assess how the prediction quality of the proposed approaches varies as the test data shifts from the domain of the training dataset. It is important to clarify that these experiments are not intended to evaluate the overall performance of the models but to gain a comparative insight into model performance in extreme cases.

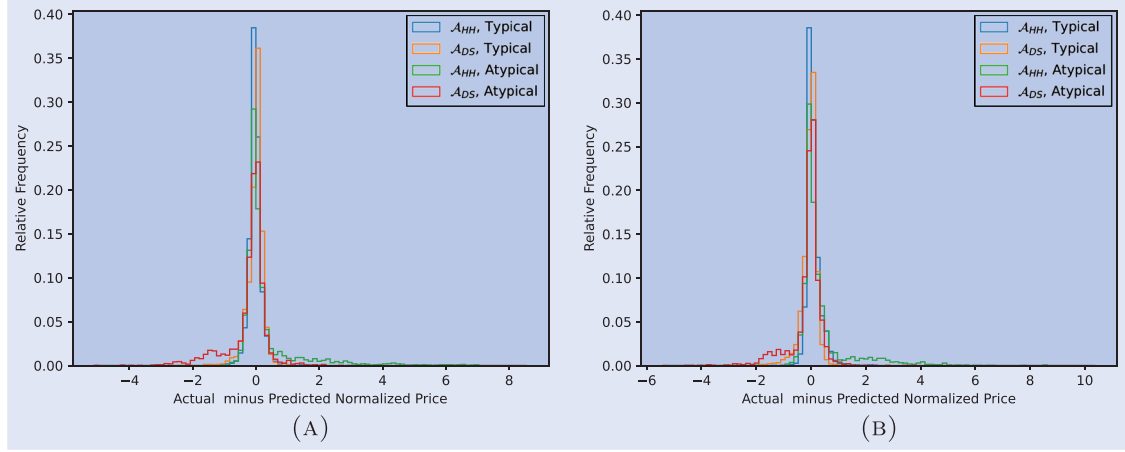


Figure 6. Histograms of subtraction of predicted from actual ‘normalized price’ are plotted for models trained on combined N50 and BNF training data. Both models with $\mathcal{A}_{HH}$, and $\mathcal{A}_{DS}$ approaches are tested on different indices with both typical and atypical scenarios. (a) NIFTY50 testing data and (b) BANKNIFTY testing data.

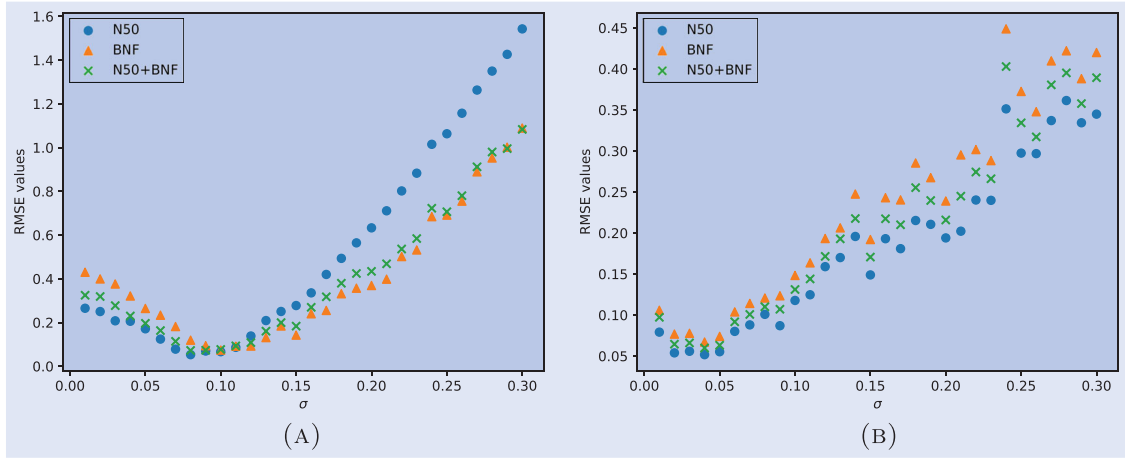


Figure 7. Model performance on a range of parametrized synthetic data. (a) Performance of $\mathcal{A}_{HH}$ models and (b) Performance of $\mathcal{A}_{DS}$ models.

These do not give a reliable performance measure in a typical scenario as both underlying asset returns and the option prices of test data are derived under specific mathematical assumptions that may not accurately reflect real-world prices.

We simulate Geometric Brownian motion (GBM) with a drift parameter $\mu = 0.1$ and vary the volatility parameter from 1% to 30% in increments of 1%. For each volatility level, we generate daily data of length 520 days. This test set is supplemented with theoretical prices of several near-ATM option contracts, with time to maturity $\{10, 25, 40\}$ and risk-free interest rate 5% calculated using the Black-Scholes-Merton formula. We then test each of the trained models for each variant of the test data and plot the predictions’ RMSE against the volatility parameter. The analysis is centered on comparing the performance of the models with $\mathcal{A}_{DS}$, $\mathcal{A}_{HH}$, and ensemble model $\mathcal{A}_{E(1,1)}$ approaches, and trained on various real datasets. Given the nature of simulated test data, although the distinction between typical and atypical test scenarios is not direct, still all the synthetic data generated with $\sigma > 20\%$ or $\sigma < 5\%$ may be viewed as atypical.

6.3.1. Experiment 1. In the first experiment, we evaluate and compare the models’ prediction quality on synthetic

test data across various training datasets. We perform this comparison for synthetic test data having varying volatility levels, and models with each approach, $\mathcal{A}_{HH}$ and $\mathcal{A}_{DS}$. Indeed, for every synthetic data with volatility parameter σ not less than 12%, among $\mathcal{A}_{HH}$ models, the BNF trained model has the least RMSE, whereas N50 trained model has the largest RMSE. This can be explained using the historical volatility (HV) values of respective training data. As the historical volatility of BNF (14.05%) is higher than that of N50 (11.38%), the test data with high σ is less unfamiliar to BNF-trained data than to N50-trained data. As noted in Goswami *et al.* (2021), the effectiveness of the approach $\mathcal{A}_{HH}$ diminishes with domain shift, i.e. the RMSE values of each $\mathcal{A}_{HH}$ model steadily increase for synthetic test data with volatility values larger than the HV of the training dataset. In fact the RMSE plots exhibit V-shaped curves, reflecting higher errors for test sets with higher mismatch in return distribution compared to the training set. On the other hand, a similar consistent comparison based on HV of the training models is not possible from the RMSE plots for the $\mathcal{A}_{DS}$ base models. Also, the growth of RMSE with higher σ values of synthetic test data is much slower for $\mathcal{A}_{DS}$ models. We recall, the $\mathcal{A}_{DS}$ approach aims to mitigate such errors. However, growth of RMSE for $\mathcal{A}_{DS}$ prediction may

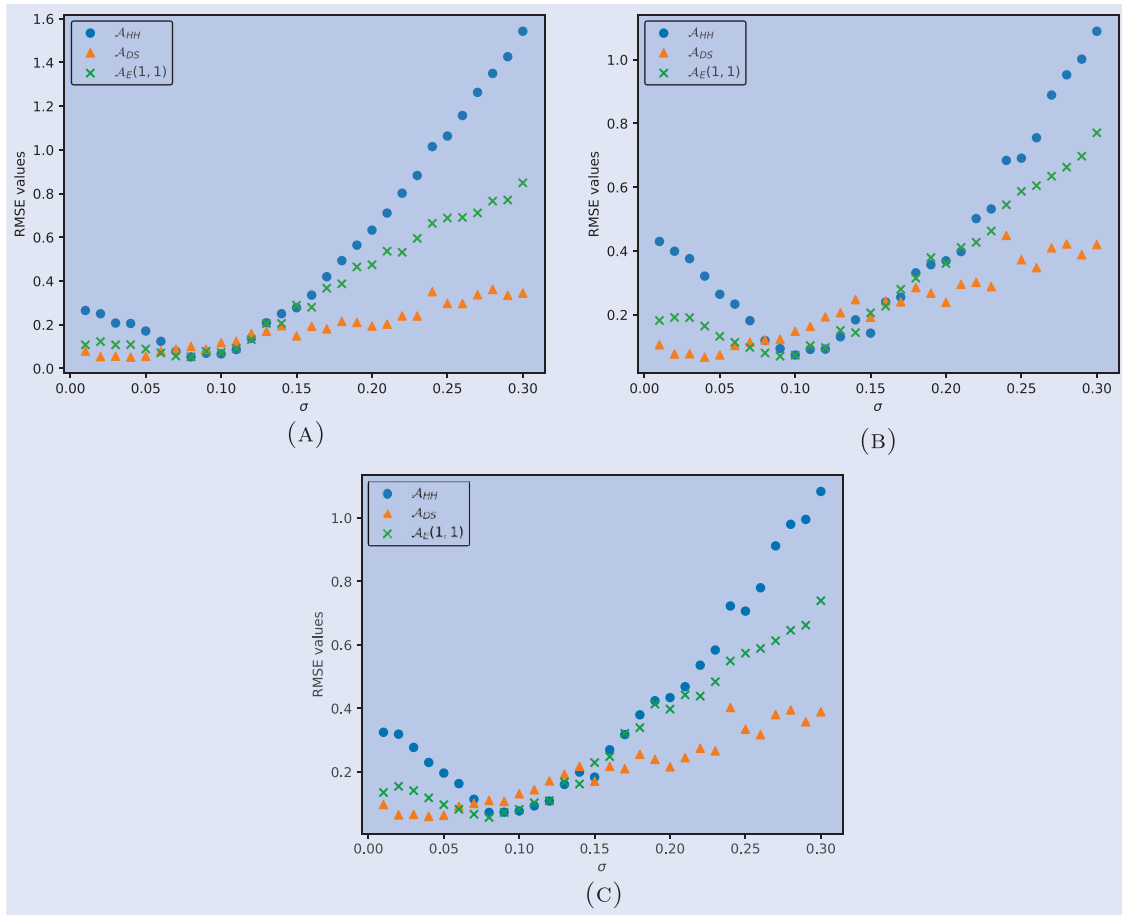


Figure 8. Comparison of approaches $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$ and $\mathcal{A}_E$ for each of three different trained set using a parametric family of synthetic data. (a) Performance of N50 trained model. (b) Performance of BNF trained model and (c) Performance of multi-source trained model.

be attributed to the growth of relative error in the approximation relation (9) with the increase of disparity (max-min ratio) of volatilities as illustrated in figure 1(b). These results align with the intent of making the $\mathcal{A}_{DS}$ approach less sensitive to discrepancies between the return distributions of the training and testing datasets. The plots also show that the RMSE of combined training is not the least but always on the lower side.

6.3.2. Experiment 2. In another experiment, we compare the RMSE values for all three approaches $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$ and the ensemble model on the aforementioned collection of synthetic data. Separate comparison plots are presented for models trained on the NIFTY50, BANKNIFTY and combined datasets. As anticipated, above a volatility level (i.e. above 17%) of the test data, the approach $\mathcal{A}_{DS}$ outperforms $\mathcal{A}_{HH}$ for every training set (see each subplot of figure 8). Similar consistent superiority is also visible for test data with σ values not more than 5%. Additionally, performance graphs for the ensemble model are presented with $\lambda_1 = \lambda_2 = 1$. In every case (i.e. subplots 8(a-c)) the RMSE of the ensemble model is mostly between the RMSEs of $\mathcal{A}_{HH}$ and $\mathcal{A}_{DS}$ models, except for a small range of the volatility values of the synthetic data, where it is smallest. However, that small range corresponds to the typical scenario, as it is in the neighborhood of the HV of the training dataset. So, it explains why table 3 conveys the best performance by $\mathcal{A}_{E(1,1)}$ models for real test datasets.

7. Conclusion

This work presents several interesting elements and fresh ideas for option pricing applications of machine learning. This paper puts forward an interesting idea of utilizing a formula for implied volatility estimates to create a common representation space so that a better domain adaptation is achieved. This method also allows training a single model on several options markets (i.e. options with different underlying securities), since this could improve out-of-sample performance in each considered market. Finally, we formulate a novel ensemble model, depending on a domain shift quotient, that works for both typical and atypical test data. Moreover, we also assess the models using a family of synthetic test data to gain a better understanding of the models. We believe this is a promising avenue for further research and could potentially bring some new insights into the field of data-driven option pricing. All the codes will be made available for academic research on request.

It will be interesting to see in future research how domain adaptation may be achieved for predicting American option prices reliably even in atypical situations. The main hurdle there will be the formulation of the common representation space because extracting implied information from American options is a hard problem. However, use of a numerical technique like Calibration Neural Network (CaNN), as formulated in Liu *et al.* (2021), is one plausible path. Further research is needed to understand this better.

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Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study. All the empirical studies reported in this article are based on openly available data.

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Appendices

Appendix 1. Derivations of theoretical results

Proof of Theorem 2.2. It is given that the market contains two risky assets, whose price processes are modeled by S_1 and S_2 , satisfying (M1)–(M3) with an auxiliary process V . For each $i = 1, 2$, let $\mathbb{P}_i$ be an equivalent martingale measure (EMM) corresponding to S_i , i.e. S_i/B is martingale under $\mathbb{P}_i$, a probability measure equivalent to $\mathbb{P}$. Using $\mathbb{P}_i$, we express a present (time t) fair price of a European-style call option on the i th risky asset having the exercise price and maturity time as K_i and T respectively, as

$$\mathbb{E}_i \left(\frac{B(t)}{B(T)} (S_i(T) - K_i)^+ \mid \mathcal{F}_t \right)$$

where $\mathbb{E}_i$ is the expectation w.r.t. $\mathbb{P}_i$. Using (M3), the above is

$$\mathbb{E}_i \left(\frac{B(t)}{B(T)} (S_i(T) - K_i)^+ \mid S_i(t), V(t) \right) = \varphi_i(t, S_i(t), V(t); K_i, T),$$

where φ_i denotes the call option price function on the i th asset with maturity T and exercise price K_i . For each $i = 1, 2$, using (1), and the above equation, and by denoting $V(t) = v$, $S_i(t) = s_i$, we get

$$\begin{aligned} & \varphi_i(t, s_i, v; K_i, T) \\ &= \mathbb{E}_i \left(\frac{B(t)}{B(T)} (S_i^{(t, s_i)}(T) - K_i)^+ \mid S_i^{(t, s_i)}(t) = s_i, V(t) = v \right) \\ &= \mathbb{E}_i \left(\frac{B(t)}{B(T)} s_i \left(S_i^{(t, 1)}(T) - \frac{K_i}{s_i} \right)^+ \mid S_i^{(t, 1)}(t) = s_i, V(t) = v \right) \\ &= s_i \mathbb{E}_i \left(\frac{B(t)}{B(T)} \left(S_i^{(t, 1)}(T) - \frac{K_i}{s_i} \right)^+ \mid S_i^{(t, 1)}(t) = 1, V(t) = v \right) \\ &= s_i \varphi_i \left(t, 1, v; \frac{K_i}{s_i}, T \right). \end{aligned} \quad (A1)$$

Furthermore, it is given that the conditional laws of $S_1^{(t, 1)}(T)$ and $S_2^{(t, 1)}(T)$ given identical V values at time t under their corresponding risk-neutral measures $\mathbb{P}_1$ and $\mathbb{P}_2$ respectively are identical. Hence for the contracts with identical time to maturity and equal moneyness at time t , i.e. $\frac{K_1}{s_1} = p = \frac{K_2}{s_2}$, we have

$$\begin{aligned} & \mathbb{E}_1 \left(\frac{B(t)}{B(T)} (S_1^{(t, 1)}(T) - p)^+ \mid S_1^{(t, 1)}(t) = 1, V(t) = v \right) \\ &= \mathbb{E}_2 \left(\frac{B(t)}{B(T)} (S_2^{(t, 1)}(T) - p)^+ \mid S_2^{(t, 1)}(t) = 1, V(t) = v \right) \end{aligned}$$

$$\text{or, } \varphi_1(t, 1, v; p, T) = \varphi_2(t, 1, v; p, T).$$

Using (A1), the above equality is rewritten as

$$s_1^{-1} \varphi_1(t, s_1, v; p s_1, T) = s_2^{-1} \varphi_2(t, s_2, v; p s_2, T).$$

Hence, (2) holds. ■

Proof of Lemma 2.5. (i) The strong solution of SDE (3) is

$$S(t) = S(0) e^{\int_0^t \left(\mu(u) - \frac{\sigma(u)^2}{2} \right) du + \int_0^t \sigma(u) dW(u)},$$

for an arbitrary initial value $S(0) > 0$. Thus S satisfies (1) and (M3) holds with no additional auxiliary process V , as S itself is Markov.

Consequently, ρ -scaling of S satisfies

$$A(t) = S(t)^{\frac{1}{\rho}} = A(0) e^{\left[\rho^{-1} \int_0^t \left(\mu(u) - \frac{\sigma(u)^2}{2} \right) du + \int_0^t \frac{\sigma(u)}{\rho} dW(u) \right]} \quad (A2)$$

as $S(0)^{\frac{1}{\rho}} = A(0)$. Therefore, A also satisfies (1) and (M3). Similarly, it is not hard to see that both S and A fulfill (M1), as an application of Girsanov's transformation and the fundamental theorem of fair pricing (See Kallianpur and Karandikar 2000, Example 2 in Subsection 7.4).

(ii) Using Itô's lemma, from (A2), we get

$$\begin{aligned} dA(t) &= \left(\frac{\mu(t)}{\rho} - \frac{\rho - 1}{2\rho^2} \sigma(t)^2 \right) A(t) dt + \frac{\sigma(t)}{\rho} A(t) dW(t) \\ \text{i.e. } \frac{1}{A(t)} dA(t) &= \left(\frac{\mu(t)}{\rho} - \frac{\rho - 1}{2\rho^2} \sigma(t)^2 \right) dt + \frac{\sigma(t)}{\rho} dW(t). \end{aligned} \quad (A3)$$

The left side is the instantaneous simple return of A . Thus the instantaneous volatility of A , i.e. the coefficient in $dW(t)$ term, is ρ^{-1} times the volatility of S . ■

Proof of Theorem 2.6. It is given that S_i solves for all $t > 0$

$$dS_i(t) = \mu_i(t) S_i(t) dt + \sigma_i(t) S_i(t) dW_i(t),$$

where the Brownian motions W_1 and W_2 may or may not be independent. Now we fix a $t > 0$ and for each $i = 1, 2$, set ρ_i arbitrarily and $A_i(u) = S_i(u)^{\frac{1}{\rho_i}}$ for all $u \geq t$ and also denote the European call option price function on the asset having price process A_i as ψ_i . In view of (A3), A_i solves the following SDE

$$dA_i(u) = A_i(u) \left(\frac{dB(u)}{B(u)} + \frac{\sigma_i(u)}{\rho_i} dW'_i(u) \right)$$

where, $W'_i(\cdot) := W_i(\cdot) + \int_0^\cdot \left(\frac{\mu_i(u)}{\sigma_i(u)} - \frac{\rho_i - 1}{2\rho_i} \sigma_i(u) \right) du - \int_0^\cdot \frac{\rho_i}{\sigma_i(u)} \frac{dB(u)}{B(u)}$. Hence

$$\frac{A_i}{B}(\cdot) = \frac{A_i}{B}(t) \mathcal{E} \left(\int_t^\cdot \frac{\sigma_i(u)}{\rho_i} dW'_i(u) \right) \quad (A4)$$

where $\mathcal{E}(X)$ denotes the Doléans-Dade exponential of the semi-martingale X . For each $i = 1, 2$, let $\mathbb{P}'_i$ denote a probability measure equivalent to $\mathbb{P}$ such that W'_i is a Brownian motion under $\mathbb{P}'_i$. This exists as a consequence of Girsanov's Theorem. Then A_i/B is a martingale under $\mathbb{P}'_i$, and hence $\mathbb{P}'_i$ is an equivalent martingale measure (EMM) corresponding to A_i .

Next, we seek for the values of constants ρ_1 and ρ_2 so that under the respective risk-neutral measures $\mathbb{P}'_1$ and $\mathbb{P}'_2$, the conditional laws of the log returns of $A_1^{(t, 1)}(T)$ and $A_2^{(t, 1)}(T)$ are identical. That is easily achieved, because of (A4), if

$$\int_t^T \frac{\sigma_1(u)}{\rho_1} dW'_1(u) \stackrel{d}{=} \int_t^T \frac{\sigma_2(u)}{\rho_2} dW'_2(u) \quad (A5)$$

where the equality ' $\stackrel{d}{=}$ ' is in the sense of distribution. Moreover, as σ_i is bounded, $\int_t^T \frac{\sigma_i(u)}{\rho_i} dW'_i(u)$ becomes Gaussian random variable with mean zero and variance $\rho_i^{-2} \mathbb{E} \int_t^T \sigma_i^2(u) du$ for each i . Hence, the values of the parameters ρ_1 and ρ_2 specified by (4) are sufficient to ensure (A5). Hence, proof of part (i) of the theorem is complete.

Since the conditional distributions of $A_i(T)/A_i(t)$ under the corresponding risk neutral measures are identical, from Lemma 2.5 and Theorem 2.2 we get (5). This completes the proof of part (ii). ■

Estimation of the volatility scalar ρ From (4), it is also evident that, if σ_1 and σ_2 are stationary, ρ_1, ρ_2 can be estimated from the past data. More precisely, ρ , the average of forward variance, see (4), is typically estimated by taking the past 20 days' data and we assume that the past 20 days' historical volatility is a good estimator of future volatility, at least in average. It is worth mentioning that ρ must be re-estimated as t and T change which is a sensible thing to do as in

Table A1. In-sample performance of XGBoost regression for $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$, and $\mathcal{A}_{E(1,1)}$ are reported.

RMSEs of in-sample option price prediction models					
Symbol(s) of Training data	Approach	Tested on N50		Tested on BNF	
		Training	Validation	Training	Validation
N50	Benchmark - BSM (HV)	0.425	0.493	0.452	0.677
	$\mathcal{A}_{HH}$	0.120	0.363	–	–
	$\mathcal{A}_{DS}$	0.150	0.306	–	–
	$\mathcal{A}_{E(1,1)}$	0.114	0.328	–	–
BNF	$\mathcal{A}_{HH}$	–	–	0.153	0.431
	$\mathcal{A}_{DS}$	–	–	0.184	0.417
	$\mathcal{A}_{E(1,1)}$	–	–	0.148	0.414
N50+BNF	$\mathcal{A}_{HH}$	0.153	0.352	0.275	0.477
	$\mathcal{A}_{DS}$	0.150	0.306	0.184	0.417
	$\mathcal{A}_{E(1,1)}$	0.140	0.320	0.223	0.447

practice one needs to calibrate the model at every epoch (i.e. as t changes and then time to maturity $T - t$ also changes).

The selection of ρ , so that (4) holds becomes particularly simple if the volatility factor σ is constant in time. Indeed, the volatility scalar ρ becomes identical to σ . Even if the volatility process is neither constant nor stationary but follows a parametric diffusion equation, the volatility scalar may be estimated by estimating the parameters. We illustrate this by considering the Heston model. In that, the square of volatility follows a Cox-Ingersoll-Ross (CIR) process, i.e. for all $t \geq 0$

$$d\sigma^2(t) = \kappa(\theta - \sigma^2(t))dt + \xi\sigma(t)d\tilde{W}(t),$$

with $\sigma^2(0) > 0$ where $\kappa, \theta, \xi, \sigma(0)$ are positive parameters and $\tilde{W}$ is a standard Brownian motion. A direct calculation shows that

$$\frac{1}{T-t} \mathbb{E} \int_t^T \sigma^2(u) du = \theta - (\theta - \sigma^2(t)) \frac{1 - e^{-\kappa(T-t)}}{\kappa(T-t)}.$$

Therefore, $\rho = \sqrt{(\theta - (\theta - \sigma^2(t)) \frac{1 - e^{-\kappa(T-t)}}{\kappa(T-t)})}$, a number between the long-run mean of volatility $\sqrt{\theta}$ and the current volatility $\sigma(t)$, can be estimated using the estimator of $\theta, \sigma(t)$ and the speed parameter κ of mean reversion.

On the other hand, for a special case of non-parametric settings, consider two stocks whose prices satisfy (3) with identical volatility processes, i.e. $\sigma_1 \stackrel{d}{=} \sigma_2$. Then (A5) holds with any equal values of ρ_1 and ρ_2 . ■

Appendix 2. Additional numerical experiments

We share outcomes of some additional numerical experiments here in this section. We first report the in-sample performance of all proposed models in table A1. The RMSE values of the testing of the models on training and validation datasets help identify the bias and variance of the models. All three proposed approaches $\mathcal{A}_{HH}$, $\mathcal{A}_{DS}$, and $\mathcal{A}_{E(1,1)}$ are investigated in table A1. For each training data and each approach, we test the resulting model for two different datasets. One is a part of the training data during 01/01/2019 – 30/04/2019, and the other is the validation data during 01/05/2019 – 31/08/2019. Moreover, we also investigate in-sample performance for the models trained with combined (both symbols) data and tested on in-sample and validation datasets of both symbols. The performance measures do not substantiate overfitting.

Next, we present the performances of a completely different set of models. In table A2, we take the benchmark as the Black Scholes

Table A2. Performance of $\mathcal{A}_{IV}(NN)$, $\mathcal{A}_{IV}$, $\mathcal{A}_{DS}$, and $\mathcal{A}_{E(1,1)}$ are assessed.

RMSEs of option price prediction models					
Symbol(s) of Training data	Approach	Test - N50		Test - BNF	
		Typical	Atypical	Typical	Atypical
BSM (IV) N50	$\mathcal{A}_{IV}(NN)$	0.169	0.411*	0.238	0.455
	$\mathcal{A}_{IV}$	0.375	1.203	0.611	1.159
	$\mathcal{A}_{DS}$	0.145	1.301	0.288	1.321
	$\mathcal{A}_{E(1,1)}$	0.242	0.537	0.284	0.442
BNF	$\mathcal{A}_{IV}(NN)$	0.140*	0.435	0.189	0.391*
	$\mathcal{A}_{IV}$	0.392	0.942	0.660	1.044
	$\mathcal{A}_{DS}$	0.153	1.199	0.206	1.215
	$\mathcal{A}_{E(1,1)}$	0.215	0.655	0.332	0.825
N50+BNF	$\mathcal{A}_{IV}(NN)$	0.152	0.468	0.185*	0.511
	$\mathcal{A}_{IV}$	0.292	0.911	0.449	0.975
	$\mathcal{A}_{DS}$	0.148	1.133	0.201	1.169
	$\mathcal{A}_{E(1,1)}$	0.188	0.628	0.244	0.623
		0.146	0.458	0.193	0.437

formula with the volatility value implied from the previously traded price of the same option. It is known to be highly accurate in predicting option prices, making further improvements difficult in a typical market scenario. We also consider another benchmark approach that is based on regression, which regresses implied volatility (IV) on option variables, like moneyness, time-to-maturity, interest rate, and $F1-F20$. Since $F1-F19$, the order statistics of centered log return represents the empirical distribution of the input data, we do not include the historical volatility (HV) as a feature. Table A2 reports the performance of XGBoost and ANN models following this approach, which we name as $\mathcal{A}_{IV}$ and $\mathcal{A}_{IV}(NN)$, respectively. Additionally, we contrasted these performances with the performance of $\mathcal{A}_{DS}$ by using IV as the estimator of the volatility scalar ρ . Finally, we have reported the performance of an ensemble model that ensembles the predictions of $\mathcal{A}_{DS}$ and $\mathcal{A}_{IV}$. In these additional experiments, it is again observed that for atypical data, $\mathcal{A}_{DS}$ performs better than $\mathcal{A}_{IV}$ and $\mathcal{A}_{IV}(NN)$, as these models devoid of domain adaptation. It is observed that the multi-source $\mathcal{A}_{E(1,1)}$ model, an unoptimized ensemble model with a simplified formula, performs better on 3 out of 4 test datasets than the benchmark. This further highlights the promise of the ensemble model that is adaptive to the domain shifts.